



SVKM'S
NMIMS[®]
Deemed to be UNIVERSITY

MPSTME

Mukesh Patel School of Technology
Management & Engineering



TRACEVEIL

AI-Assisted Real-Time Smart Home IoT
Digital Forensics and Incident Response Platform

Group 18

Ayra Jha — K028

Atharva Rangnekar — K053

Aarya Rao — K054

Faculty Mentor: Prof. Rohit Suryawanshi



Introduction

Smart-home environments generate evidence across sensors, devices, network activity and system records, but these sources do not naturally share one common forensic representation.



One investigation model for **heterogeneous IoT evidence**.

Problem Statement

Smart-home forensic evidence is fragmented across devices, vendors and data formats, making consistent investigation and cross-source correlation difficult.



Heterogeneity

Different IoT ecosystems expose different data structures, schemas and representations.



Ring

{ ... }



Philips

< ... >



Investigation Complexity

Fragmented records, timestamp inconsistencies and source-specific formats complicate event reconstruction and correlation.



10:01:12



10:00:47



10:01:31



AI Reliability

Generative AI can assist investigators, but probabilistic output should not become the authority for forensic scores, alerts or evidence.



AI Assistant
(supportive only)

Suggestions,
summaries, hypotheses



Forensic
Analysis

Deterministic,
validated evidence

--- Not in forensic decision boundary ---



RESEARCH QUESTION

How can heterogeneous historical and live IoT evidence be unified into a reproducible forensic workflow while preserving provenance and keeping AI outside the forensic decision boundary?

Literature Survey – Key Findings

- 1



Heterogeneous Smart-Home Forensics
Shin et al., 2024

Highlights investigation difficulty across multiple vendor ecosystems, proprietary storage formats, and clock/timestamp differences.
- 2



Common Smart-Home Architecture
Kim et al., 2025

Suggests a generalized common architecture and staged forensic methodology for scalable smart-home investigations.

- 3



IoT-CAD
2025

Combines heterogeneous IoT traces with AI-based attack detection and attribution; useful comparison point for Traceveil's deterministic investigator-facing workflow.

- 4



LLMs in Digital Forensics / ForensicLLM
Research area

Shows promise for forensic assistance and natural-language interaction, while raising hallucination, reproducibility, and confidentiality concerns.

- 5



Public Evidence Sources
Evaluation data

CASAS smart-home telemetry and TON_IoT device telemetry are the primary public evaluation sources; controlled simulation supplements them.



Overall Insight: The surveyed work highlights the need to unify heterogeneous smart-home evidence and support reproducible, investigator-friendly workflows.

Research Gap

Among the systems and literature reviewed for this project, we did not identify one that combines all four of the following in the **same investigation workflow**:

01**Batch + Live Evidence**

Historical datasets and controlled live-device telemetry

**02****Canonical Representation**

One provenance-preserving event model

**03****Deterministic Investigation**

Rules, risk scoring, event correlation, timelines and graph views

**04****Bounded Local AI**

Evidence-grounded explanation without forensic authority



Traceveil targets this integration gap.

Project Objectives & Scope

Objectives



1

Validate IoT evidence and preserve integrity and provenance.



2

Normalize heterogeneous batch and live inputs into the versioned Canonical Event Model.



3

Apply deterministic detection rules, behavioural baselines and bounded risk scoring.



4

Correlate related events and reconstruct investigation timelines and device/entity relationships.



5

Present validated investigation results through case-scoped APIs and dashboard views.



6

Provide optional local, evidence-grounded AI assistance.

Scope



In Scope:

Case and evidence management; CASAS and TON_IoT telemetry; controlled simulation; controlled live telemetry; deterministic analysis; timelines; graphs; reports; local AI assistance.



Out of Scope:

Production-scale IoT monitoring; unauthorized interception or device access; automated containment; autonomous attack execution; legal-admissibility claims; replacement of enterprise SIEM/IoT-security platforms.

Why Traceveil?

Different tools, different design goals.
Traceveil is built for a focused smart-home digital-forensics workflow.



Enterprise tools are broader. Traceveil is narrower by design — and better suited to reproducible smart-home IoT forensic investigation.

Its advantage is the combination of offline evidence, live telemetry, provenance-preserving normalization, deterministic analysis and local AI within one focused workflow.

Tools, Technologies & Datasets



Frontend

User Interface & Visualisation



React



Vite



TypeScript



Tailwind CSS



Recharts



XYFlow



Backend & APIs

Core Services & Data APIs



Python 3.11+



FastAPI



Pydantic



Analysis & Storage

Data Processing & Persistence



Pandas



NetworkX



SQLite



Local AI

On-Device Inference & Models



Ollama



Qwen 9B



Live Communication

Real-time Data Exchange



Authenticated
HTTP Telemetry



Server-Sent
Events (SSE)



Reports & Testing

Documentation, Testing & API Contracts



ReportLab



Pytest



Vitest



OpenAPI-generated
TypeScript types



Controlled IoT Integration

Integration Support (Laboratory Setup)



ESP32 reference
implementation



controlled laboratory
telemetry

Supports integration and development; not indicative of final physical deployment.



Datasets & Evaluation Sources



Primary Evaluation Sources

Main datasets for evaluation and benchmarking



CASAS Smart Home Dataset



TON_IoT Device Telemetry



Supplemental

Additional data for controlled evaluation



Controlled Simulation



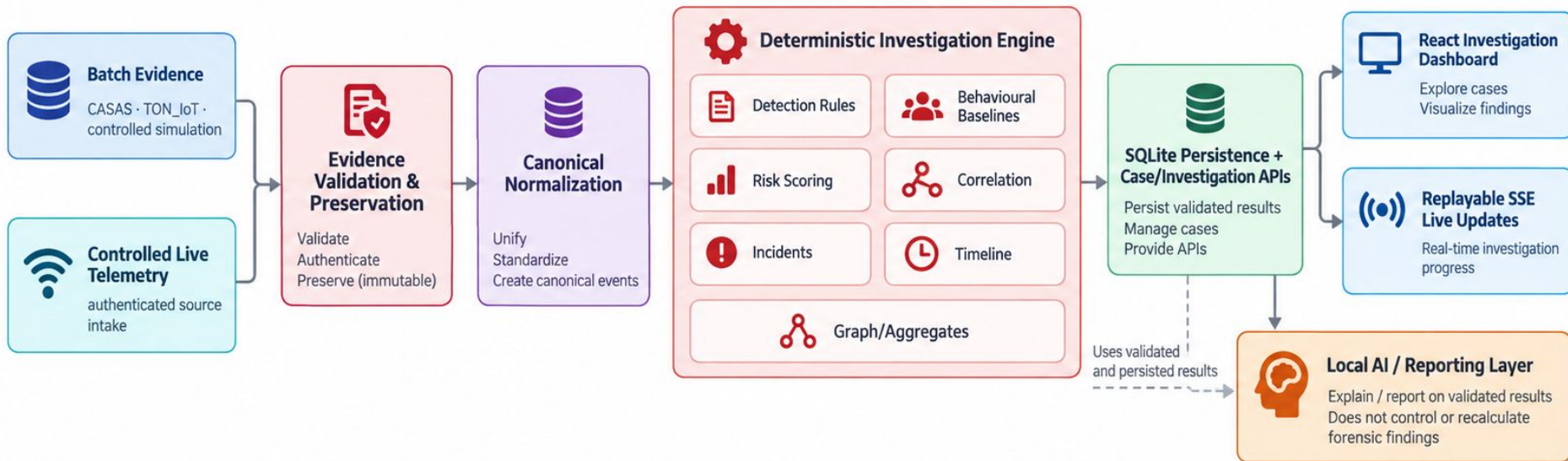
Secondary Compatibility

For cross-dataset compatibility only



CICIoT2023 Network-Flow Data

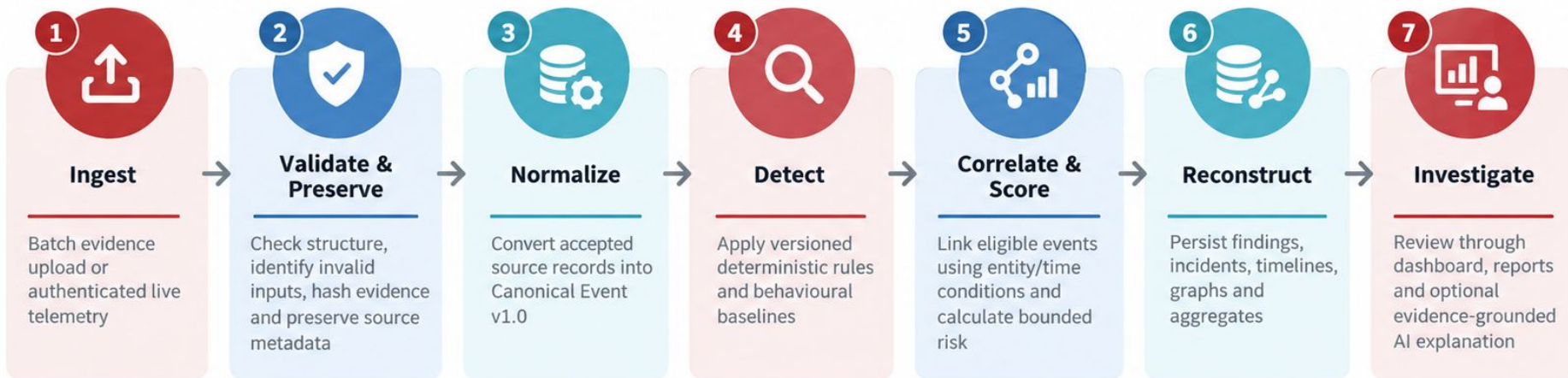
System Architecture



Batch and accepted live evidence become canonical events before forensic findings are calculated.

Methodology — Evidence to Investigation

FROM RAW EVIDENCE TO ACTIONABLE INSIGHTS



Canonical Event

A normalized, consistent representation of an observed event.



Event Object

Canonical Schema

#	Event ID	: <event_id>
📁	Case ID	: <case_id>
📅	Observed timestamp	: <observed_timestamp>
🕒	Backend ingestion timestamp	: <ingestion_timestamp>
📄	Event type	: <event_type>
👤	Device / Actor / Target	: <device actor target>
🔗	Network or event attributes	: <network_or_event_attributes>



Provenance

Forensic Lineage

🔖	Evidence ID	: <evidence_id>
🗄️	Batch or live origin	: <batch_or_live_origin>
📡	Source type and source ID	: <source_type> / <source_id>
🖐️	Source hash	: <source_hash>
📄	Raw-record hash	: <raw_record_hash>
⚙️	Adapter version	: <adapter_version>
🔗	Normalization version	: <normalization_version>

Deterministic Analysis

Rule-based detection for consistent, explainable forensic outcomes.



Implemented Rule Families

Deterministic • Repeatable



LABEL-001

source-labelled malicious activity



RATE-001

requests at or above 3× the declared baseline



AUTH-001

10 authentication failures within 60 seconds for a stable actor/target relationship



BASELINE-001

statistical behavioural-baseline spike



Risk Score Composition

Weighted Factors



- Severity — 30%
- Confidence — 25%
- Repetition — 20%
- Device Criticality — 15%
- Corroboration — 10%



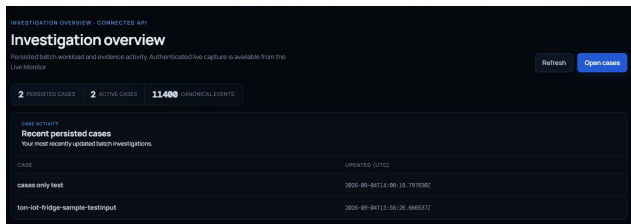
Forensic Authority Mandate — Python calculates forensic facts; AI may explain persisted results but cannot create or recalculate scores, alerts or evidence.

Implementation & Current Progress

Progress as per the formal project roadmap and reconciliation.

✓ Completed (7/12 major areas)

- ✓ Initialization
- ✓ Frontend & Investigation Interface
- ✓ Evidence Validation
- ✓ Canonical Event Model & Normalization
- ✓ Live IoT Integration
- ✓ Filtering, Detection, Risk & Correlation
- ✓ Dashboard Results & Investigation APIs



65% Complete 35% Remaining



Continuing development towards a complete end-to-end demonstration and project delivery.

○ Remaining (5/12 major areas)

- Real-Time Event Delivery
- Visualization Engine + Qwen
- AI Investigation Chat
- Alerts, Email, Reports & Automation
- Physical Live Demonstration

Showing 1-50 of 5000 persisted records

EVENT ID	OBSERVED AT	EVENT TYPE	DETAILS
76058f52-56e9-5442-900f-d37c3d72fab	2009-10-16T00:01:04.000098Z	device_state	Inspect
22a8e788-fc5d-5421-8f0c-c7ab60999522	2009-10-16T03:55:53.000080Z	device_state	Inspect
734c568f-5e69-52a7-a006-d14b5eb40c82	2009-10-16T06:00:51.000068Z	device_state	Inspect
23b08606-eb09-5a25-93b4-c0625619471d	2009-10-16T06:15:19.000078Z	device_state	Inspect



Frontend tests
Passing



Backend/API/Persistence tests
Passing



TypeScript/Lint
Passing



Production Build
Passing

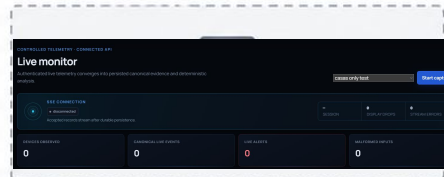
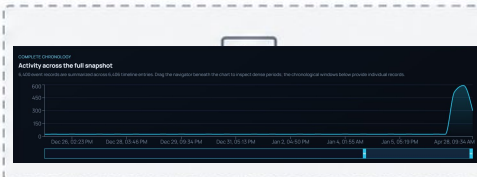
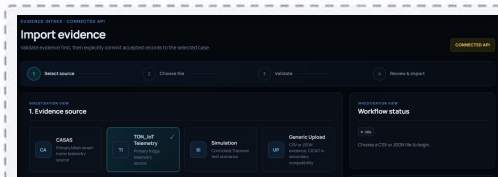
Live Demonstration

BATCH INVESTIGATION WORKFLOW



LIVE APPLICATION DEMO

CONTROLLED LIVE PATH



**Recorded Walkthrough —
Backup**



*Demonstration focuses on the application workflow.
Physical-device demonstration shown only if assembled and validated.*

Roadmap Ahead

Current Formal Roadmap Status — **65% Complete** | 93 / 143 sub-items verified



Beyond Current Capstone Scope

Broader IoT datasets & network conditions • Multi-user authentication/authorization • Scalable external storage/job infrastructure

Conclusion



Traceveil brings heterogeneous historical and controlled live smart-home IoT evidence into a single provenance-preserving forensic workflow, where deterministic backend logic performs detection, risk scoring, correlation and reconstruction, while local AI is limited to assisting the investigator rather than deciding forensic facts.



65% Formal Roadmap Completion — 93 / 143 planned sub-items verified as of 6 September 2026.



65%



Next: complete the remaining real-time, AI/reporting and physical demonstration milestones.

References

- [1] “TON_IoT dataset,” *IEEE Access*, 2020.
- [2] Washington State University, “CASAS Smart Home Datasets.”
- [3] “IoT-CAD,” *Ad Hoc Networks*, 2025.
- [4] D.-H. Shin et al., “heterogeneous IoT incident investigations,” *Applied Sciences*, 2024.
- [5] S. Kim et al., “common architecture-based smart-home forensic investigation,” 2025.
- [6] “Large Language Models in Digital Forensics: Capabilities, Challenges and Future Directions,” 2026.
- [7] “ForensicLLM: A Local Large Language Model for Digital Forensics,” 2025.
- [8] NIST SP 800-86, “Guide to Integrating Forensic Techniques into Incident Response.”
- [9] NIST SP 800-92, “Guide to Computer Security Log Management.”

THANK YOU

Traceveil